

Advanced (Carolina Reaper)

- Q1.** What is the graph of $y = |x|$?
- (A) A straight line
(B) A parabola
(C) A V-shaped graph
(D) A circle
(E) An ellipse
- Q2.** The vertex of $y = |x - 2|$ is at
- (A) (0, 0)
(B) (2, 0)
(C) (0, 2)
(D) (2, 2)
(E) (1, 1)
- Q3.** What is the axis of symmetry of $y = |x + 1|$?
- (A) $x = 0$
(B) $x = 1$
(C) $x = -1$
(D) $y = 0$
(E) $y = 1$
- Q4.** The graph of $y = 2|x|$ opens
- (A) Upward
(B) Downward
(C) To the left
(D) To the right
(E) Not at all
- Q5.** The transformation of $y = |x|$ to $y = |x| + 3$ is a
- (A) Horizontal shift
(B) Vertical shift
(C) Horizontal stretch
(D) Vertical stretch
(E) Reflection
- Q6.** The graph of $y = |x - 1| - 2$ has a vertex at
- (A) (0, 0)
(B) (1, -2)
(C) (0, 2)
(D) (2, 2)
(E) (1, 1)
- Q7.** What is the equation of the axis of symmetry of $y = |x - 3| + 1$?
- (A) $x = 0$
(B) $x = 1$
(C) $x = 3$
(D) $y = 0$
(E) $y = 1$
- Q8.** The graph of $y = -|x|$ opens
- (A) Upward
(B) Downward
(C) To the left
(D) To the right
(E) Not at all

Open Ended Questions (FRQ)

- Q9.** Graph $y = |x + 2| - 1$ and identify the vertex, axis of symmetry, and direction of opening. Required space: 5 lines
- Q10.** Find the equation of the absolute value function with a vertex at $(-2, 1)$ and an axis of symmetry at $x = -2$. Required space: 5 lines

Chef's Tips

- Tip 1: Remind students to show all work for free response questions.
- Tip 2: Encourage students to check their graphs for accuracy.
- Tip 3: Suggest students use graph paper to help with graphing.